

Chapter 9 Homework Answer Sheet

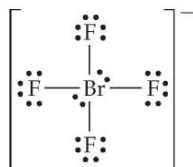
Part A Multiple choice (15 points)

1-5 EBCEE 6-10 CBAAD 11-15 CBEED

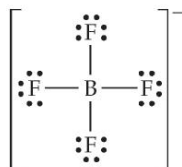
Part B Short questions (20 points)

16. (7 points)

(a) (4 points)



6 e^- pairs around Br
octahedral e^- domain geometry
square planar molecular geometry



4 e^- pairs around B,
tetrahedral e^- domain geometry
tetrahedral molecular geometry

Even though both ions have four bonding electron domains, the six total electron domains around Br require octahedral electron-domain geometry and square planar molecular geometry, whereas the four total electron domains around B lead to tetrahedral electron-domain and molecular geometries.

(b) (3 points)

All molecules in the series have tetrahedral electron-domain geometry and bent molecular geometry. The less electronegative the central atom, the larger the nonbonding electron domain, the greater the effect of repulsive forces on adjacent bonding domains. The angles will vary as $H_2O > H_2S > H_2Se$.

17. (3 points)

(a) (iii) **(b) sp^3**

18. (4 points)

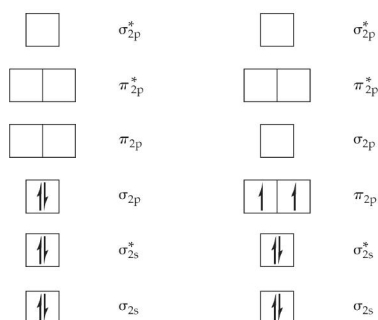
(a) (2 points) a bonding σ MO

(b) (1 point) 2

(c) (1 point) (iv) longer and weaker

19. (6 points)

(a) (3 points)



The two possible molecular orbital electron configurations of B_2 are shown above. If the σ_{2p} MO is lower in energy than the π_{2p} MOs, B_2 would have no unpaired electrons and be diamagnetic. If the π_{2p} MOs are lower in energy than the σ_{2p} MO, B_2 has two unpaired electrons and is paramagnetic.

(b) (3 points) According to the Molecular Orbital Theory, the two highest-energy electrons of O_2 are in antibonding π^*_{2p} MOs and O_2 has a bond order of 2.0. Removing these two electrons to form O_2^{2+} produces an ion with bond order 3.0. O_2^{2+} has a stronger O–O bond than O_2 , because O_2^{2+} has a greater bond order.